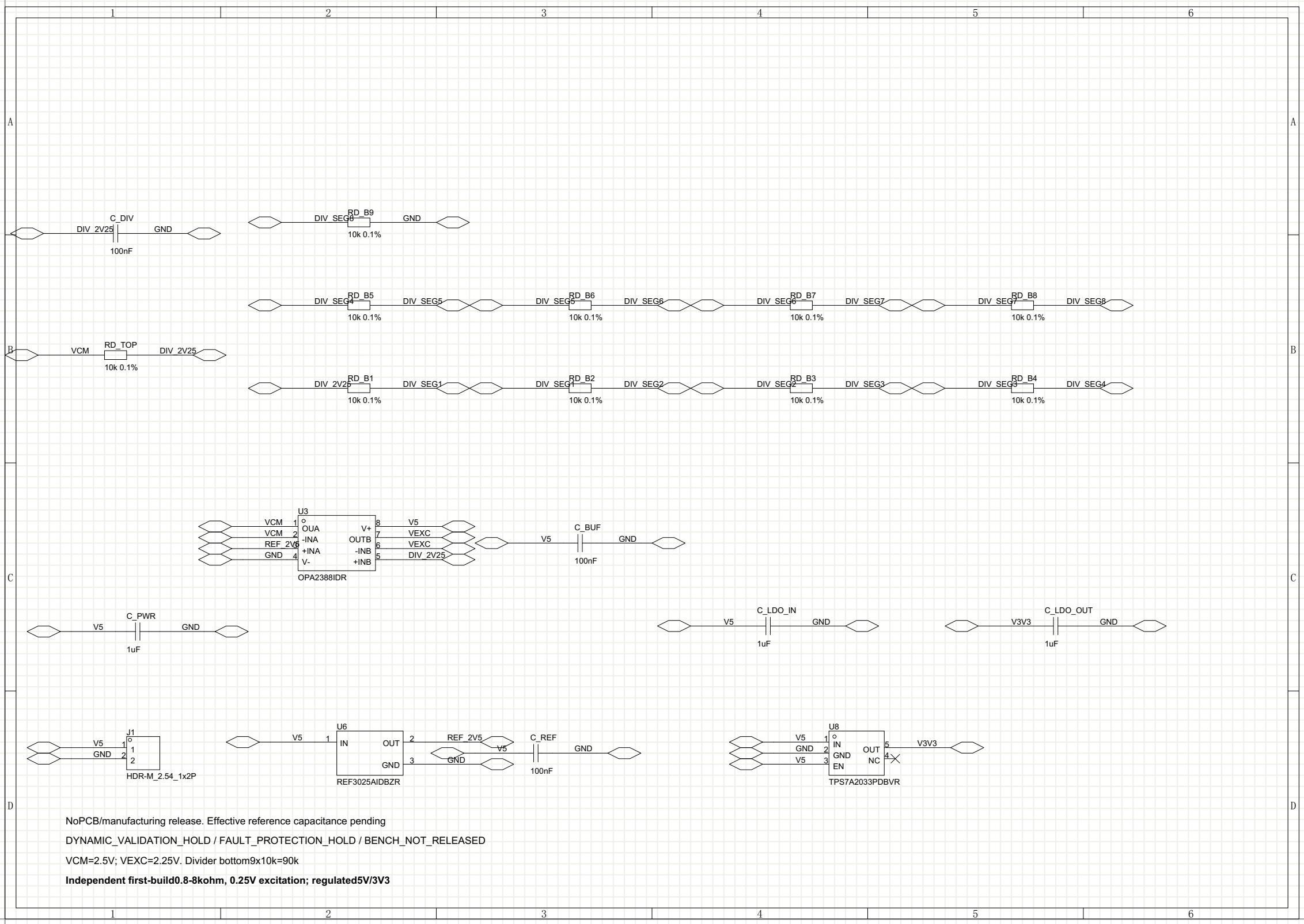
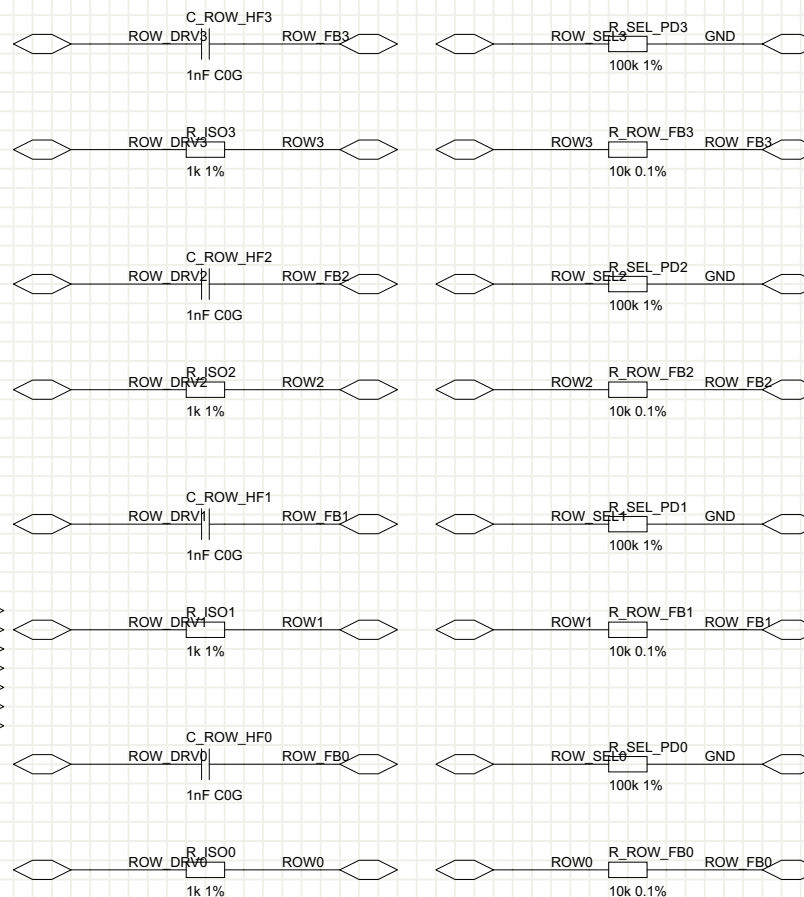
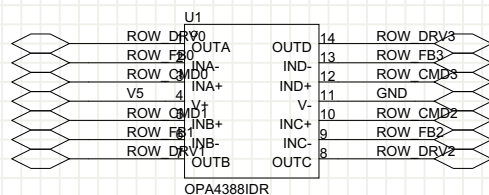
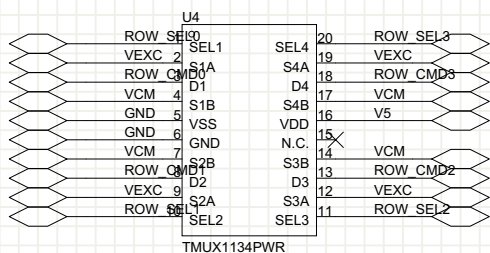
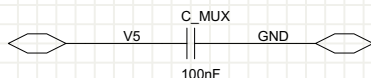
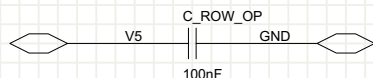
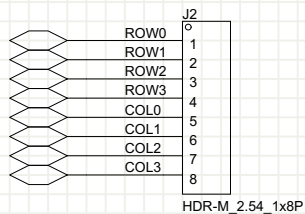




NoPCB/manufacturing release. Effective reference capacitance pending
DYNAMIC_VALIDATION_HOLD / FAULT_PROTECTION_HOLD / BENCH_NOT_RELEASED
VCM=2.5V; VEXC=2.25V. Divider bottom9x10k=90k
Independent first-build0.8kohm, 0.25V excitation; regulated5V/3V3

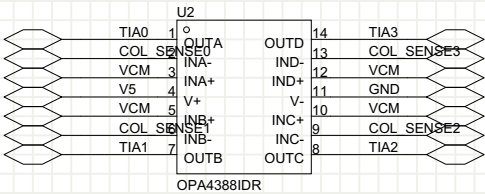
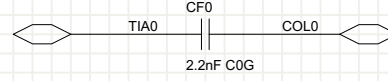
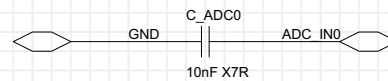
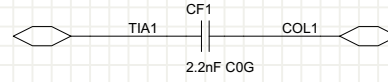
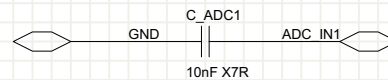
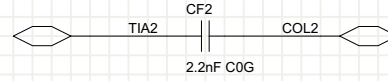
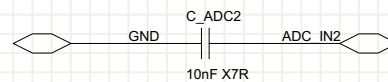
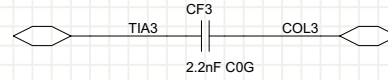
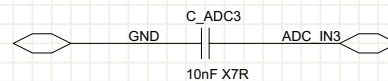
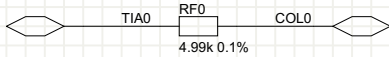
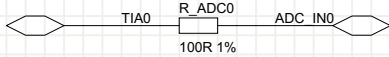
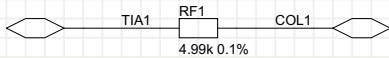
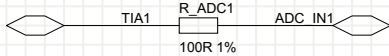
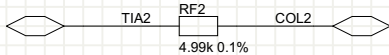
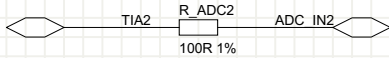
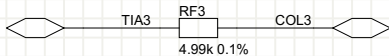
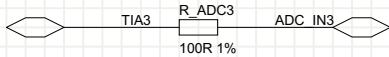
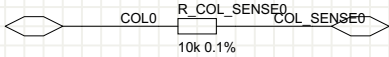
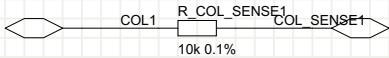
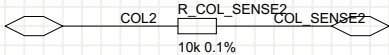
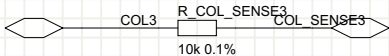
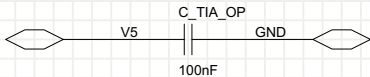


GPIO reset/HiZ SEL pulldowns; fault safe-state/10ms detection not bench verified

J2:1..4 ROW0..3;5..8 COL0..3; external fixed8-electrode array

Row ISO changed 10R -> 1k. Feedback senses board ROW after resistor

TMUX1134:SEL0 -> B=VCM (idle);SEL1 -> A=VEXC (active)



DYNAMIC_VALIDATION_HOLD: no vendor macro simulation executed

External lead/contact resistance is not canceled by board-end feedback

10k sense resistors added at opamp input; sensor current flows through RF/CF

Four TIA feedback loops always closed; VCM to all noninverting inputs

A

A

B

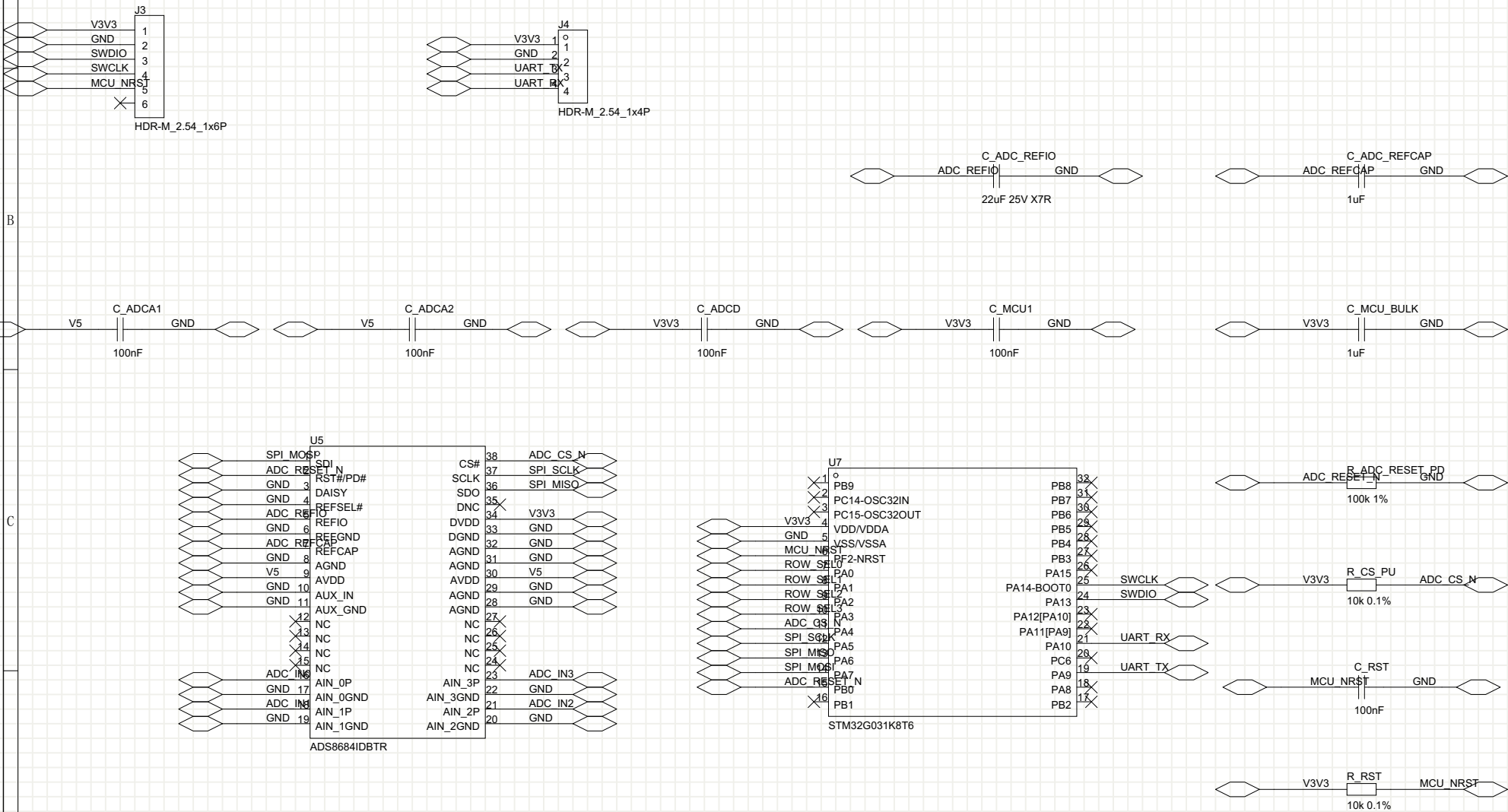
B

C

C

D

D



FAULT_PROTECTION_HOLD: supply collapse/backfeed/recovery not closed
100fps target;20kSPS total;4MHz SPI; pipelined channel labels must be checked
AIN0/1=16/18;AIN2/3=21/23. AllAIN_GND toGND, notVCM
ADS8684 internal4.096V reference; REFSEL/DAISY low;35DNC strictly NC